

From Waves to Words: A Neuro-Symbolic Approach to Personalized Health Monitoring

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Abstract

The basic idea is that the large language models are unable to interpret the physiological time series data directly. and it is also a difficult task for a human to interpret this. The Vitals ai agent can ingest, compiles and convert the raw physiological time series data into a structured format that is helpful for large language models as well as the human medical practitioner. we have implemented human activity recognition and stress prediction in the vitals ai agent. details about har details about stress prediction details about the architecture of the vitals ai agent evaluation of the vitals ai agent evaluations steps results of the evaluation discussion of the results conclusion and future work. Abstract here [?].

Keywords:

1. Introduction
2. Related Work
3. Experiment
4. Methodology
5. Runtime Trace and Evidence Boundary
6. Discussion
7. Conclusion

Declarations

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Ethics. No new clinical deployment or primary human-subject data collection was conducted. The reported runtime instantiation uses secondary human research datasets under the original providers’ access terms and should be interpreted as a non-clinical pre-deployment architecture study.

Competing interests. The authors declare no competing interests.

AI assistance. OpenAI Codex (GPT-5.5-xhigh) was used during manuscript revision to assist with LaTeX editing and wording audits. AI-assisted tools were not used to select, screen, or code the literature corpus. The authors reviewed and edited the AI-assisted outputs and take full responsibility for the content of this publication.

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Data availability. No new clinical or primary human-subject datasets were generated. Capture24 and WESAD are secondary human research datasets available subject to the original dataset providers’ access terms. The repository contains the LaTeX source, root `main.pdf`, and an artifact-status manifest; executable runtime artifacts are not included in this paper-source repository and are reserved for a separate artifact package.

References